
Report Title: Concise Technical Subtitle

Course, Project, Laboratory, or Conference Track

CentraleSupélec, Department or Research Group

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Abstract

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Introduction

Use the introduction to move from the broad context to the precise contribution of the report. The first paragraph should identify the practical or scientific problem and explain why it is worth studying now. The second paragraph should define the scope of the work, including the system, data, experiment, corpus, or case study considered. The final paragraph should state the report’s main objective and outline the organization of the remaining sections.

When adapting this template, replace generic claims with specific evidence and cite sources where readers would expect support. Use non-breaking spaces before cross-references and citations; the code examples below show the source command and the rendered result.

Figure`\ref{fig:sample_diagram}` renders as Figure 1.
 Table`\ref{tab:inputs}` renders as Table 1.
`\cite{Lamport1994}` renders as [1].

Keep the introduction readable for a technically informed audience that may not share your exact specialization. If the work follows an assignment, laboratory, or project brief, describe the intellectual task rather than copying the prompt.

This sentence shows how to add a footnote with a project page, repository, or personal website¹.

1 Background and Related Work

Use this section for the background, definitions, standards, or previous work needed to read the report. The sample citations here demonstrate bibliography usage and should be replaced by sources that are actually relevant to the project [1, 2, 3].

Keep citations close to the statement they support. For example, write that a method is standard [2] in the relevant sentence rather than placing all references at the end of a section. Add bibliography entries to `references.bib`, and keep all entries in one consistent citation style and level of detail.

2 Data, Materials, or Case Description

Use this section to present inputs, materials, cases, or notation that need a compact reference. Table 1 illustrates the expected pattern: introduce the table in the text, place `\label{...}` immediately after `\caption{...}`, include units, and discuss the table after it appears.

Table 1: Template table for reporting inputs, parameters, or experimental conditions.

Item	Description	Value or Range	Note
Input A	State what this variable represents.	10 m	Source or short note.
Input B	Include units in the value or header.	1 s to 5 s	Formatting note.
Input C	Mark estimates clearly.	Estimated	Short explanatory note.

Tables should have captions that allow them to stand on their own. Put units in headers, align comparable numbers, and avoid excessive precision. Use `booktabs` rules as in Table 1; avoid vertical rules unless they genuinely improve readability. If a table is too wide, reconsider whether all columns are necessary before reducing the font size.

¹Available at <https://heitor-gama.com>.

3 Methods

Use the methods section for notation, equations, workflows, or procedures. Define notation before using it, keep variable names consistent, and cite a source when using a standard method.

$$M = \frac{1}{n} \sum_{i=1}^n w_i x_i, \quad (1)$$

where M is the reported metric, x_i is an observation or component, w_i is a weight, and n is the number of observations. Replace Equation 1 with the central equation of the report and define every symbol immediately after the equation.

Displayed equations are part of the surrounding sentence and should carry the punctuation required by that sentence. In Equation 1, the comma after the formula is intentional because the sentence continues with “where”. If the sentence ends at the equation, place a period at the end of the displayed formula.

3.1 Algorithm or Workflow

Use a short algorithm description when implementation details affect the report. Prefer pseudocode that explains the logic rather than language-specific code. Use code in a particular programming language only when the point is to show that the implementation is simple or when syntax itself matters. Place longer code or configuration details in the appendix or a repository, and reference as in the main text, as in Appendix A.

```

1 inputs: values x_i and weights w_i
2 total_weight <- 0
3 weighted_sum <- 0
4
5 for each pair (x_i, w_i) in the inputs:
6     total_weight <- total_weight + w_i
7     weighted_sum <- weighted_sum + w_i * x_i
8
9 return weighted_sum / total_weight

```

4 Results

Use this section to place figures and tables close to the text that discusses them. Introduce each visual before it appears, then refer to it again when explaining what the reader should notice.

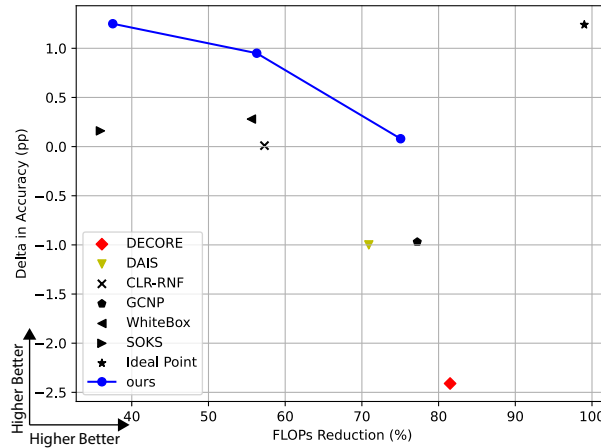


Figure 1: Example vector figure. The caption states what is shown and why the visual encoding remains readable without relying only on color.

Every figure and every table should be referenced in the text, as done here with Figure 1 and Table 1. Prefer vector outputs such as PDF for plots, diagrams, and schematics; reserve raster formats such as PNG or JPEG for photographs or image data. Keep all figure text in the same language as the report; if a source figure is in another language, reconstruct or redraw it from the original source when necessary. Always label plot axes and include units directly in the axis labels, for example t (s) or power (MW). A hard title inside a plot can usually be omitted because the caption identifies the figure; add one only when it helps, such as when several subplots need short internal titles. Use color carefully: a colorblind reader should still understand the plot, so combine color with line style, markers, labels, direct annotations, or texture. Do not use red/green contrast as the only signal.

5 Discussion

Use the discussion section when it improves the structure of the report. Keep paragraphs short enough to scan, refer back to figures, tables, and equations instead of repeating them, and separate this section from the conclusion if both are included.

6 Limitations and Future Work

Use this section when limitations or next steps would otherwise make the conclusion too long. Keep limitations specific and traceable to the report, and format future work as a short set of concrete directions rather than an open-ended wish list.

Conclusion

Use the conclusion for the final short section of the report. Avoid introducing new figures, tables, acronyms, or unexplained references here unless the course instructions require them.

Acknowledgements

Use this optional section to thank supervisors, instructors, collaborators, technical staff, funding sources, or people who provided feedback but are not listed as authors. Keep acknowledgements concise and check whether the course or venue expects them to be included, omitted, or anonymized.

Use of AI Tools

This report may have used AI tools for limited support tasks such as drafting, editing, code review, formatting, literature organization, or checking consistency. The authors remain responsible for the accuracy of the technical content, the validity of the analysis, the interpretation of results, and the proper citation of all sources. Replace this paragraph with the disclosure required by your course, venue, or institution.

References

- [1] Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, 2 edition, 1994.
- [2] Frank Mittelbach, Michel Goossens, Johannes Braams, David Carlisle, and Chris Rowley. *The LaTeX Companion*. Addison-Wesley, 2 edition, 2004.
- [3] Edward R. Tufte. *The Visual Display of Quantitative Information*. Graphics Press, 2 edition, 2001.

Appendix

A Supplementary Material

Use appendices for material that supports reproducibility or auditability but would interrupt the main argument. Examples include long derivations, extended tables, additional figures, survey instruments, data dictionaries, parameter files, or implementation notes.

Table 2: Template data dictionary or symbol table.

Field or Symbol	Meaning
x_i	Observation, sample, design variable, or measured quantity.
w_i	Weight, coefficient, or confidence factor associated with x_i .
M	Aggregate metric reported in the main text.

B Reproducibility Checklist

Before submitting, replace this checklist with course-specific requirements or remove it. Common formatting checks include: every figure and table is referenced, captions are readable, units are explicit, sources are cited, and the AI disclosure is adjusted to the rules of the course.

B.1 Checklist Notes

Use subsections in the appendix when they group several related details under a clear topic. This heading is numbered as B.1 because it is the first subsection of Appendix B.

B.1.1 Appendix Heading Granularity

Appendix subsubsections are useful for long derivations, implementation details, or repeated experiment variants. Do not split the appendix into many tiny headings just to make it look comprehensive; readers use the appendix to audit details, so its organization should remain compact and searchable.